

# Exercises: LINQ

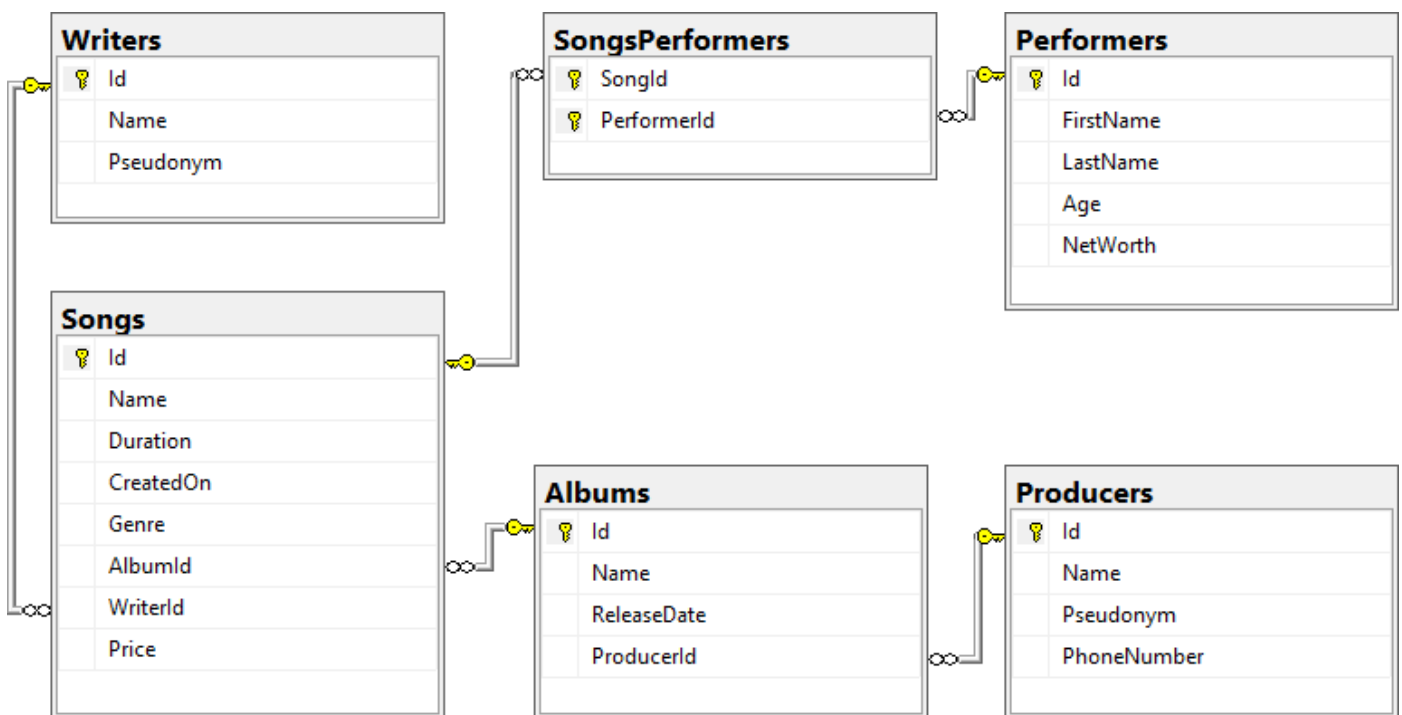
You can check your solutions in [Judge](#)

## MusicHub

People love listening to music, but they see that YouTube is getting older and older. You want to make people happy and you've decided to make a better version of YouTube – **MusicHub**. It's time for you to start coding. Good luck and impress us.

### 1. MusicHub Database

You must create a **database** for a **MusicHub**. It should look like this:



### Constraints

Your **namespaces** should be:

- **MusicHub** – for your **Startup** class, if you have one
- **MusicHub.Data** – for your **DbContext**
- **MusicHub.Data.Models** – for your **Models**

Your **models** should be:

#### Song

- **Id** – integer, **Primary Key**
- **Name** – text with **max length 20** (required)
- **Duration** – **TimeSpan** (required)
- **CreatedOn** – date (required)
- **Genre** – genre **enumeration** with possible values: "**Blues, Rap, PopMusic, Rock, Jazz**" (required)

- **AlbumId** – integer, Foreign key
- **Album** – the Song's Album
- **WriterId** – integer, Foreign key (required)
- **Writer** – the Song's Writer
- **Price** – decimal (required)
- **SongPerformers** – a collection of type **SongPerformer**

## Album

- **Id** – integer, Primary Key
- **Name** – text with max length 40 (required)
- **ReleaseDate** – date (required)
- **Price** – calculated property (the sum of all song prices in the album)
- **ProducerId** – integer, foreign key
- **Producer** – the Album's Producer
- **Songs** – a collection of all **Songs** in the **Album**

## Performer

- **Id** – integer, Primary Key
- **FirstName** – text with max length 20 (required)
- **LastName** – text with max length 20 (required)
- **Age** – integer (required)
- **NetWorth** – decimal (required)
- **PerformerSongs** – a collection of type **SongPerformer**

## Producer

- **Id** – integer, Primary Key
- **Name** – text with max length 30 (required)
- **Pseudonym** – text
- **PhoneNumber** – text
- **Albums** – a collection of type **Album**

## Writer

- **Id** – integer, Primary Key
- **Name** – text with max length 20 (required)
- **Pseudonym** – text
- **Songs** – a collection of type **Song**

## SongPerformer

- **SongId** – integer, Primary Key
- **Song** – the performer's Song (required)
- **PerformerId** – integer, Primary Key
- **Performer** – the Song's Performer (required)

## Table relations

- **One Song** can have **many Performers**
- **One Performer** can have **many Songs**

- **One Writer** can have **many Songs**
- **One Album** can have **many Songs**
- **One Producer** can have **many Albums**

**NOTE:** You will need a constructor, accepting **DbContextOptions** to test your solution in **Judge!**

## 2. All Albums Produced by Given Producer

You need to write method **string ExportAlbumsInfo(MusicHubDbContext context, int producerId)** in the **StartUp** class that receives a **ProducerId**. Export **all albums** which are **produced by** the provided **ProducerId**. For each **Album**, get the **Name**, **ReleaseDate** in format the "MM/dd/yyyy", **ProducerName**, the **Album Songs** with each **Song Name**, **Price** (formatted to the second digit) and the **Song WriterName**. **Sort the Songs by Song Name (descending)** and by **Writer (ascending)**. At the end export the **Total Album Price** with exactly **two digits after the decimal place**. **Sort the Albums by their Total Price (descending)**.

### Example

| Output (producerId = 9)                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-AlbumName: Devil's advocate -ReleaseDate: 07/21/2018 -ProducerName: Evgeni Dimitrov -Songs: ---#1 ---SongName: Numb ---Price: 13.99 ---Writer: Kara-lynn Sharpous ---#2 ---SongName: Ibuprofen ---Price: 26.50 ---Writer: Stanford Daykin -AlbumPrice: 40.49 ...</pre> |

## 3. Songs Above Given Duration

You need to write method **string ExportSongsAboveDuration(MusicHubDbContext context, int duration)** in the **StartUp** class that receives **Song** duration (**integer, in seconds**). Export the songs which are **above** the given duration. For each **Song**, export its **Name**, **Performer Full Name**, **Writer Name**, **Album Producer** and **Duration (in format("c"))**. **Sort the Songs by their Name (ascending)**, and then by **Writer (ascending)**. If a **Song** has more than one **Performer**, export all performers and sort them (**ascending, alphabetically**). If there are no **Performers** for a given song, don't print the "**---Performer**" line at all.

### Example

| Output (duration = 4)                                                                   |
|-----------------------------------------------------------------------------------------|
| <pre>-Song #1 ---SongName: Away ---Writer: Norina Renihan ---Performer: Lula Zuan</pre> |

```
---AlbumProducer: Georgi Milkov
---Duration: 00:05:35
-Song #2
---SongName: Bentasil
---Writer: Mik Jonathan
---Performer: Zabrina Amor
---AlbumProducer: Dobromir Slavchev
---Duration: 00:04:03
-Song #3
---SongName: Carvedilol
---Writer: Chloe Trayhorn
---Performer: Rhody Bettam
---Performer: Tine Althorp
---AlbumProducer: Evtim Miloshev
---Duration: 00:02:39
...
```